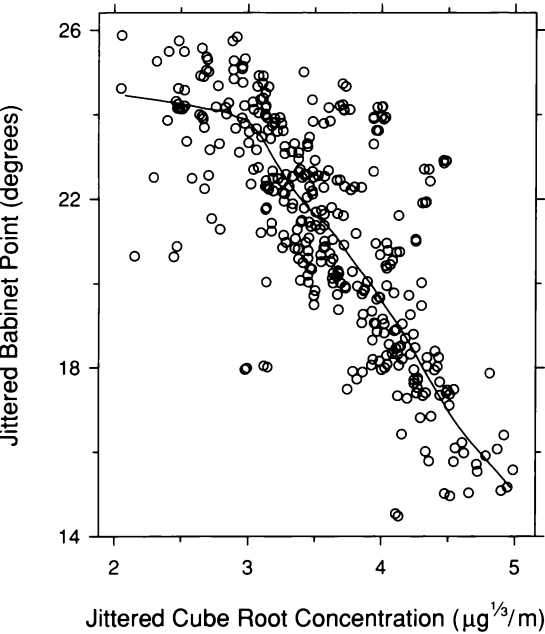
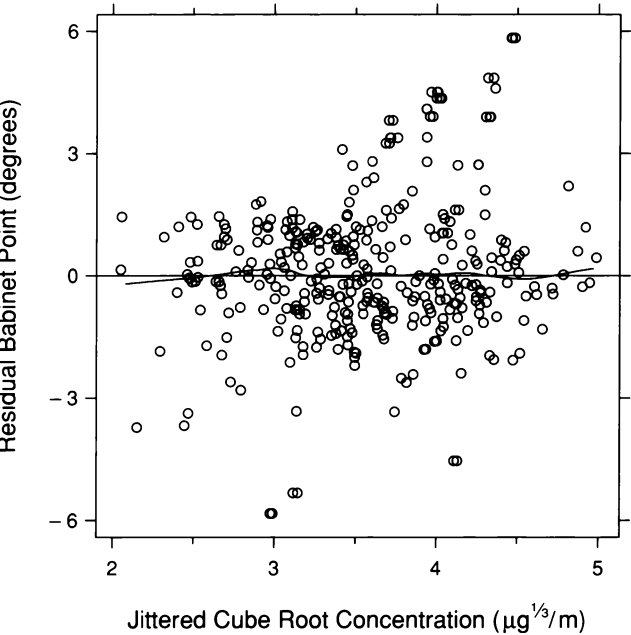
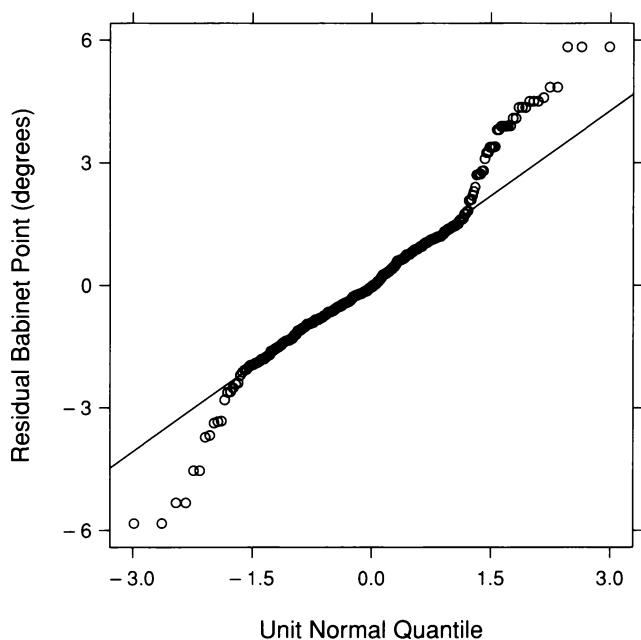




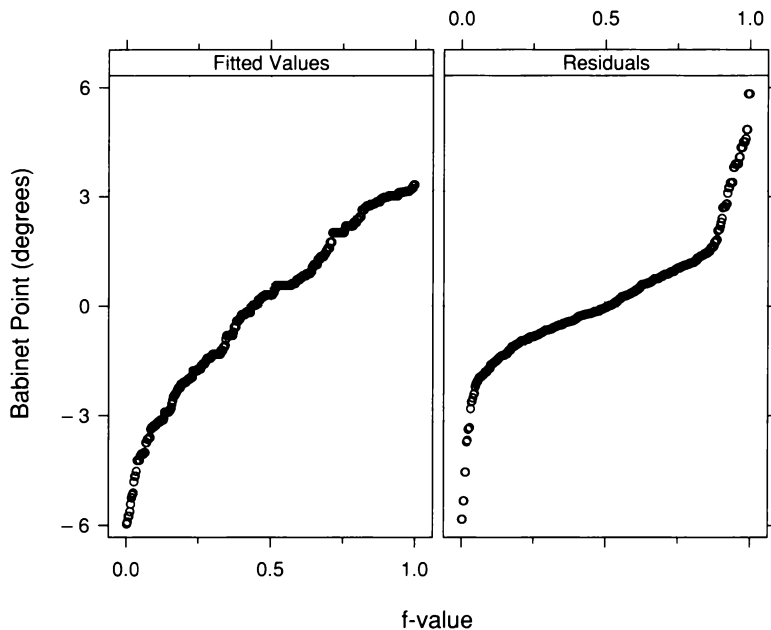
3.36 The curve is a robust loess fit to the polarization data with $\alpha = 1/3$ and $\lambda = 1$. The curve is banked to 45°.



3.37 The residuals from the third robust loess fit are graphed against cube root concentration. The parameters of the robust loess curve on the plot are $\alpha = 1/3$ and $\lambda = 1$.



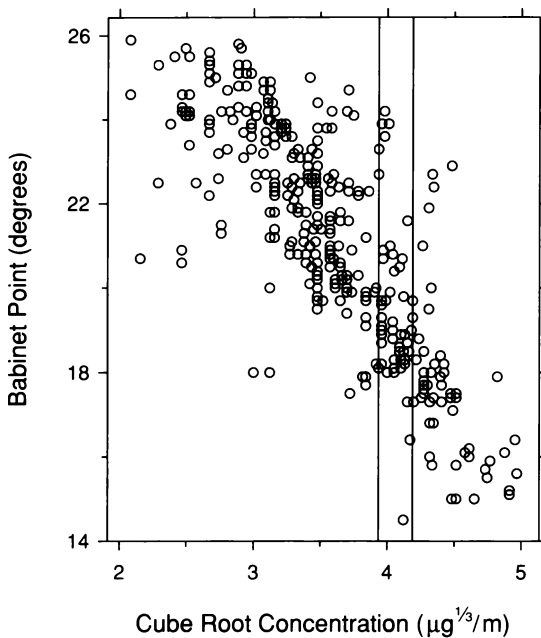
3.38 A normal q-q plot displays the distribution of the residuals from the final robust loess fit to the polarization data.



3.39 An r-f spread plot compares the spreads of the residuals and the fitted values minus their mean for the final robust loess fit to the polarization data.

3.7 Slicing

The full purpose in analyzing factor-response data is to visualize the change in the distribution of the measurements of the response as the factor changes. This is illustrated in Figure 3.40 by the polarization data. The vertical slice contains observations whose values of c , the cube root particulate concentration, lie in a small interval. For these observations, the values of b , the Babinet point, are a set of univariate data with a distribution. We can pick many such slices and compare the resulting distributions to see how the distribution of b changes as c changes. The robust loess curve fitted to these data in Section 3.5 shows the change in the location of b with c , but there are other aspects of the distribution to consider.

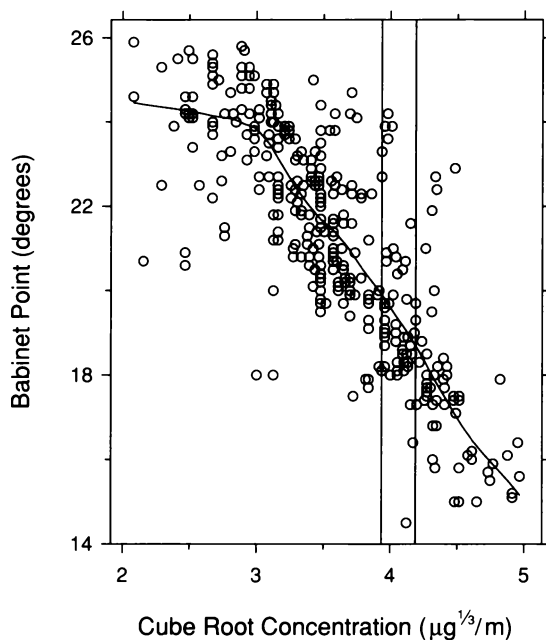


3.40 The display illustrates slicing, a method for studying the conditional distribution of the response given the factor.

Stated in this way, the goal of the analysis of the polarization data becomes that for univariate data, studied in Chapter 2. For example, distributions of singer heights for different voice parts were compared to study how height depends on voice part. For the polarization data, we will compare distributions of values of b for different slices to study how b depends on c .

Visualizing Residual Distributions

In Chapter 2, distributions were visualized by a number of tools such as box plots and q-q plots. We could apply these directly to distributions of b formed from slices on c . But there is a problem that is illustrated by Figure 3.41, which displays the polarization data and the final robust loess fit of Section 3.6. Within the slice, the location of the distribution of b given c is not constant. Thus, the values of b within the slice are not homogeneous. Were we to compare with another slice where the location is constant but the spread is the same, say around $c = 2.5\sqrt[3]{\mu\text{g}/\text{m}}$, the distribution for the first slice might look as if it had greater spread simply because the location is changing. The change in location interferes with a study of other aspects of the distribution of b given c .

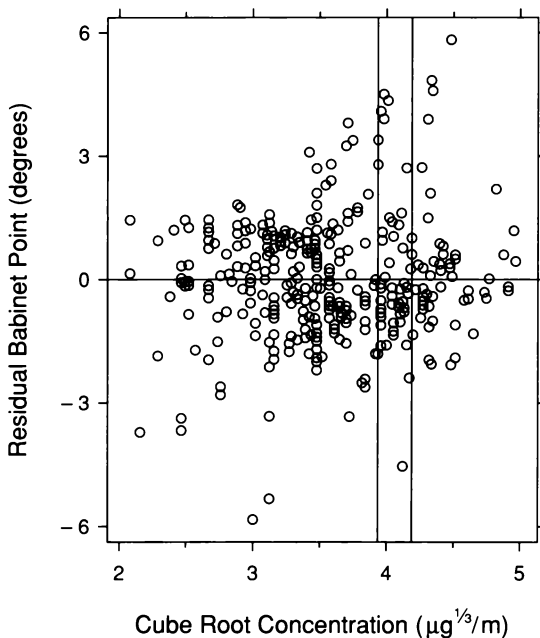


3.41 The robust loess curve describes how the location of the distribution of the response changes with the factor.

One remedy to the problem of a changing location would be to condition on very small intervals so that the change within each slice is small. But the result would be many groups of measurements and only a

few measurements within each group. There is a better solution: carrying out the analysis with residuals from the robust loess fit. This is illustrated with Figure 3.42. With the fit removed, the location is now constant, not just within the slice but overall since we have determined from the residual dependence plot in Figure 3.37 that there is no remaining dependence of the location of b on c . In removing the change in the location, the hope is that the remaining aspects of the distribution change less precipitously so that we can use wider slices.

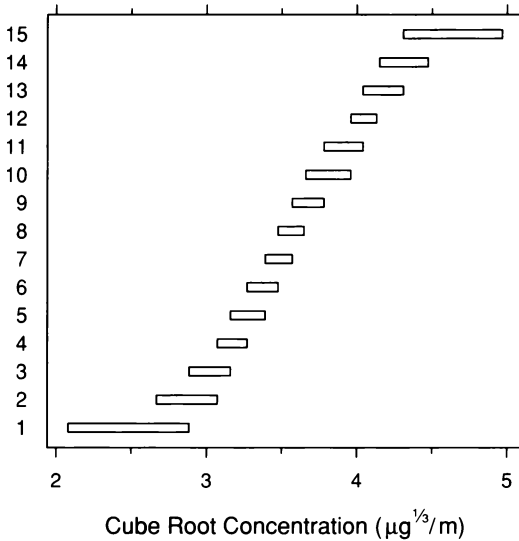
Just from Figure 3.42 itself there is a suggestion that aspects of the distribution of b given c other than location might be changing. For example, in the interval of values of c from about $3.5 \sqrt[3]{\mu\text{g}}/\text{m}$ to $4.5 \sqrt[3]{\mu\text{g}}/\text{m}$, the upper tail of the residual distribution appears stretched out. Slicing will provide more information about this effect.



3.42 The display illustrates slicing the residuals to study how aspects of the distribution of the response other than location change with the factor.

Slicing Intervals

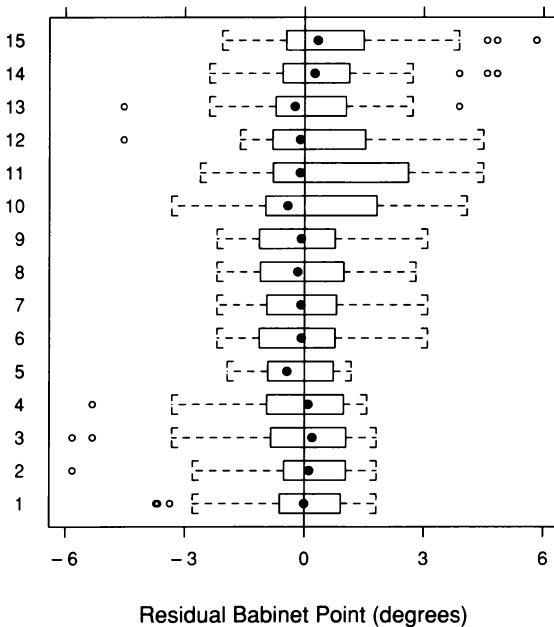
Figure 3.43, a *given plot*, shows intervals of c that we will use for slicing. The intervals are output from the *equal-count algorithm*. The input is the number of intervals and the target fraction of values shared by successive intervals. For Figure 3.43, the number of intervals is 15 and the target fraction is $1/2$. The algorithm takes the input and selects intervals to make the numbers of values in the intervals as nearly constant as possible, and to make the fraction of values shared by each pair of successive intervals as nearly equal to the target fraction as possible. The lower boundary of the lowest interval is the minimum value of the measurements of c , and the upper boundary of the highest interval is the maximum value of the measurements. The algorithm, which is described at the end of this section, does its best, but achieving equality is typically not possible. For example, for the intervals in Figure 3.43, the number of values of c ranges from 45 to 61. This variation occurs in large part because there are many ties among the measurements of c ; this can be seen in Figure 3.42.



3.43 The given plot shows 15 intervals selected by the equal-count method. The target fraction of overlap is $1/2$.

Sliced Distribution Plots

Figure 3.44 shows box plots of the 15 sets of residuals formed by the slicing intervals of the given plot in Figure 3.43. As we go from bottom to top through the box plots, the corresponding intervals go from bottom to top on the given plot. The box plots reveal a change in the distribution of the residuals. As the intervals of c increase, there is a migration from a stretching out of the lower tail to a stretching out of the upper tail. Thus there is substantial nonhomogeneity in the distribution of the residuals. The ideal case, as in Chapter 2, is a location fit that produces homogeneous residuals; then we can pool them to produce a more powerful description of variation about the fit. For example, for the singer heights in Chapter 2, once the means of the voice parts are subtracted, the residual distribution no longer depends on voice part; the eight residual distributions are homogeneous and can be pooled. For the analysis of the ganglion data in previous sections of this chapter, the residuals for the quadratic fit to CP ratio produced nonhomogeneous residuals; their spread increased with the location. But a log transformation resulted in a fit with homogeneous residuals. For the



3.44 Fifteen conditional distributions of residual Babinet point are compared by box plots.

polarization data, the residuals are not homogeneous, but in this case, there is no simple remedy such as transformation to make them so. Thus we cannot pool all of the residuals to characterize the behavior of the distribution at each value of c . The distribution at c must be characterized by residuals corresponding to cube root concentrations close to c , for example, by the box plots in Figure 3.44. Sometimes the results of a data analysis are unavoidably complicated.

Choosing the Slicing Intervals

The two parameters of the equal-count algorithm used in slicing typically need to be chosen by trial and error. There is no one correct solution; different choices can show different properties. Still, there are considerations that affect the choice.

Suppose the number of observations is n , the number of intervals is k , and the target fraction of overlap is f . As shown in a coming section that is for the record, the target value for the number of observations in each interval is

$$r = \frac{n}{k(1 - f) + f}.$$

The value of r , which is not necessarily an integer, increases as k decreases or as f increases.

The numbers of values in the intervals must be controlled to be neither too small nor too large. If they are too small, there is no basis for comparison. If they are too large, there is a risk that within an interval there is a significant change in the distribution that is masked by too much pooling.

Unless there is an enormous amount of data, it is preferable to have the intervals overlap. The slicing intervals then move gradually along the scale of the factor, which provides greater sensitivity in detecting nonhomogeneity than would be provided by intervals that do not overlap and thus change less gradually. This same principle is used in loess fitting. For the fit at a single value of x , a vertical slice contains the observations that receive positive weight in the fitting; since the loess fit is evaluated typically at a grid of x values with far smaller spacing than

the widths of the slices, the fit is determined by a sequence of slices that slide along the factor scale. In loess fitting, the widths of the slices are controlled by the choice of the smoothing parameter, α . In slicing to compare residual distributions, the widths of the slices are controlled by the choices of the parameters k and f .

For the Record: The Equal-Count Algorithm

Let x_i be the values of the factor, ordered from smallest to largest, so x_1 is the smallest and x_n is the largest. The equal-count algorithm finds intervals whose endpoints are values of x_i . Let ℓ_j be the lower endpoint of the j th interval, and let u_j be the upper.

Consider, first, a special case with three properties: (1) there are no ties among the x_i ; (2) there are intervals for which the actual fractions of values shared by successive intervals are all equal to f ; (3) the numbers of values in these intervals are equal to a constant, r . The total number of values in all of the intervals collectively is the number of conditioning intervals, k , times the number of values, r , in each interval. But also, since two successive intervals share fr values, and since there are $(k - 1)$ separate sets of such shared values, the total number of values is $n + (k - 1)fr$. Thus

$$rk = n + (k - 1)fr$$

and

$$r = \frac{n}{k(1 - f) + f}.$$

Now we can compute the interval endpoints. For the first interval,

$$\ell_1 = x_1$$

and

$$u_1 = x_r.$$

For the second interval,

$$\ell_2 = x_{1+(1-f)r}$$

and

$$u_2 = x_{r+(1-f)r}.$$

The formulas for the endpoints of the j th interval are

$$\ell_j = x_{1+(j-1)(1-f)r}$$

and

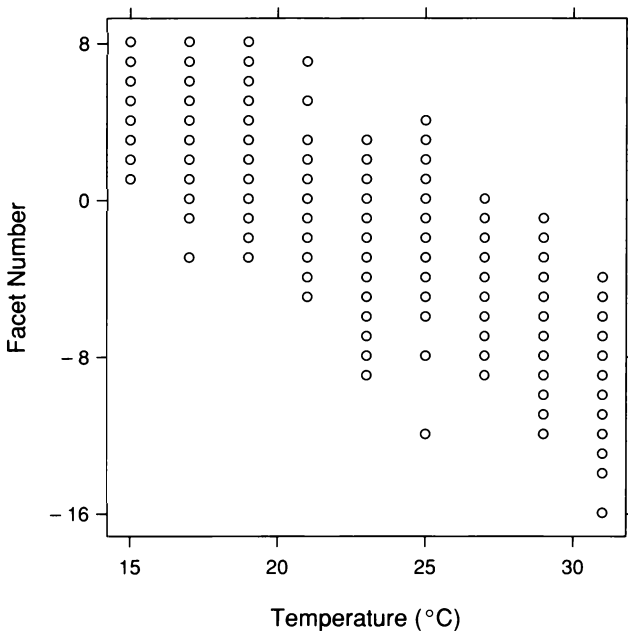
$$u_j = x_{r+(j-1)(1-f)r}.$$

In the general case, we cannot necessarily make the actual fractions exactly equal to f , or make the numbers of points in the intervals constant, because the indices that define ℓ_j and u_j in the above formulas are not necessarily integers. In the general case, we simply compute the values $1 + (j - 1)(1 - f)r$ and $r + (j - 1)(1 - f)r$, and round them to the nearest integer to get indices. In all cases, $\ell_1 = x_1$ and $u_n = x_n$, so the first interval has a left endpoint of x_1 and the last interval has a right endpoint of x_n .

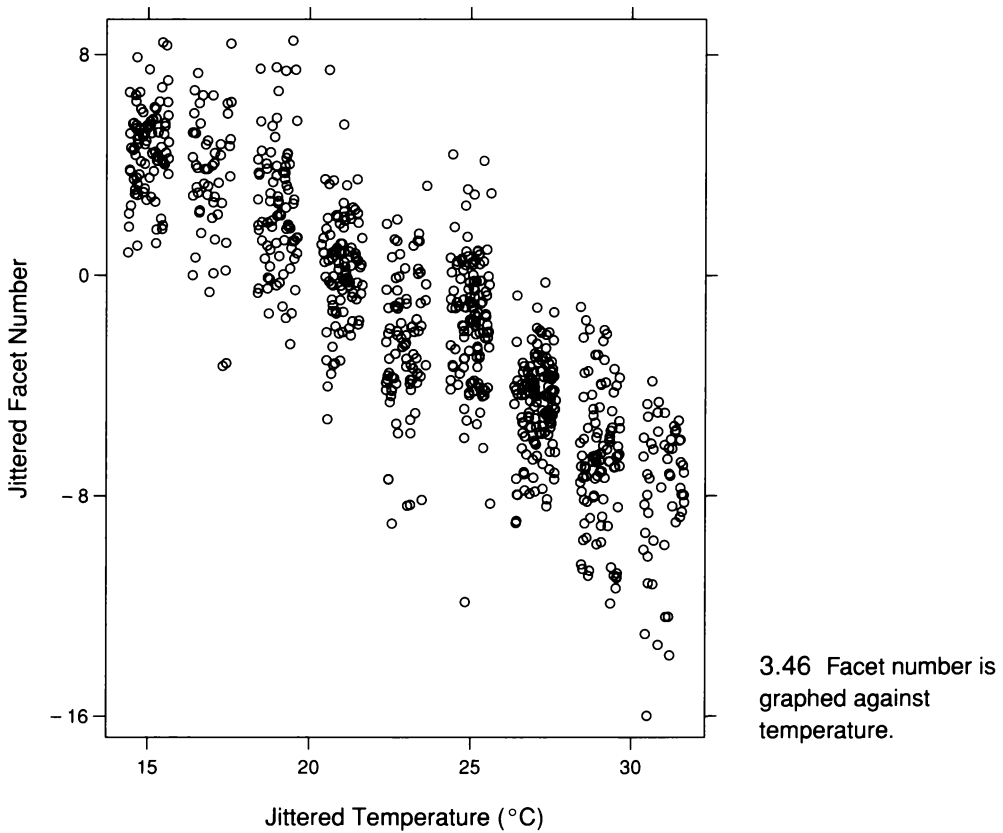
3.8 Discrete Values of a Factor

In 1924, a journal article reported 823 observations from a genetics experiment on flies' eyes [50]. Stocks of the ubiquitous species *Drosophila melanogaster* Meig were hatched in nine incubators whose temperatures varied from 15°C to 31°C in equal steps of 2°C. The number of facets of the eyes of each hatched fly were reported in units that essentially make the measurement scale logarithmic. The goal of the experiment was to see how facet number depends on temperature.

In Figure 3.45, the facet numbers are graphed against the temperatures; most plotting symbols are visually lost because of the repeat measuring at each temperature and the rounding of the facet numbers. In Figure 3.46, the data are graphed with temperature and facet number jittered to alleviate the overlap.



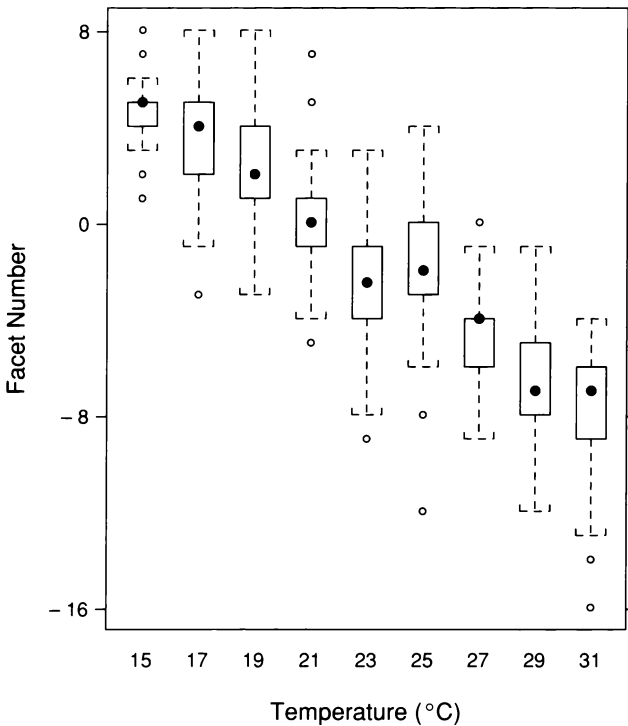
3.45 Facet number is graphed against temperature.



In the fly experiment, the measurements of the factor are discrete; there are many measurements of the response for each unique value of the factor. The data at the outset are ready for slicing. The facet numbers at each temperature serve as a single slice. Since there is no variation of the factor within each slice, there is no problem of a changing location within a slice, and we can begin the data analysis with slicing rather than waiting until we have residuals from a location fit.

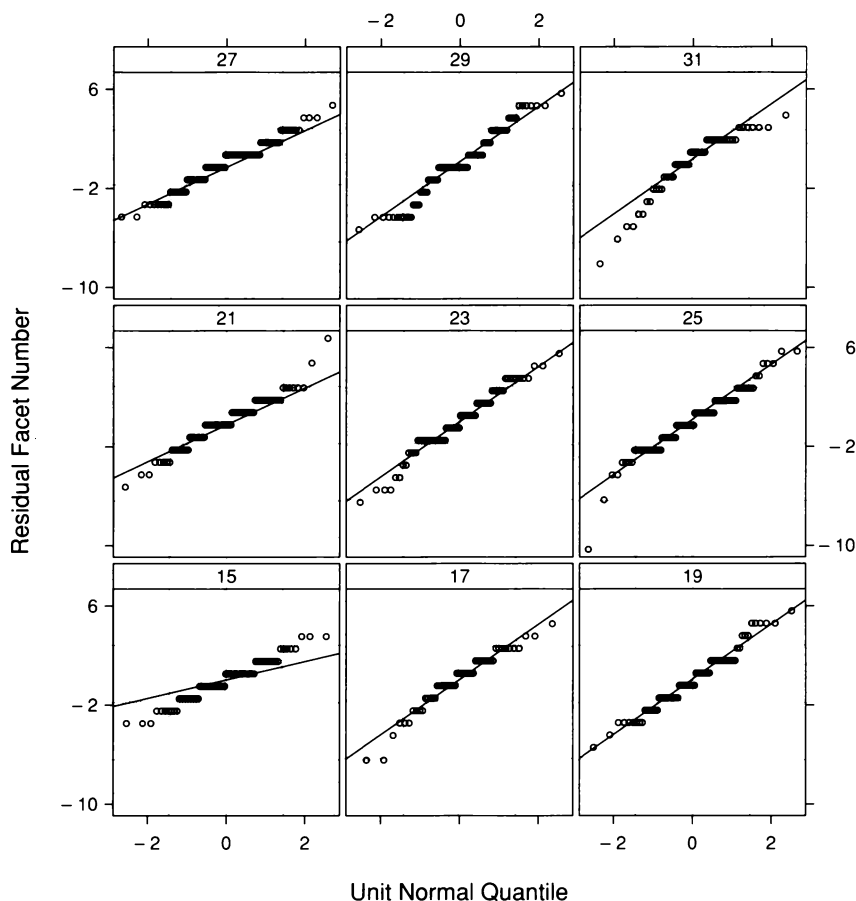
Figure 3.47 graphs the nine distributions of facet number by box plots. The overall impression is that the shifts in the distributions are additive, varying exclusively in location, and that the distributions are symmetric. Furthermore, the shifts are large compared with the spreads of the distributions, which means that temperature is an important factor in determining facet number.

Figure 3.48 shows normal q-q plots, not of the facet numbers themselves, but of the residuals from the means because this provides a better look at the normal approximation. If we graph the facet numbers, the large change in location forces the data on each panel into a narrow interval along the vertical scale, which interferes with our ability to assess the shape of the underlying pattern of the points. The nine residual distributions appear well approximated by the normal. The biggest deviation, that for the largest temperature, is mild skewness.



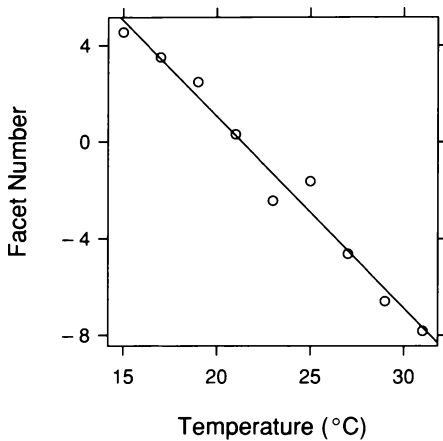
3.47 The conditional distributions of facet number given temperature are compared by box plots.

For 15°C, the underlying pattern is linear but the quartile line does not follow the pattern because facet number at 15°C takes on only eight distinct values and about half of the data are equal to the two central values. The slopes of the underlying patterns for the nine temperatures are quite similar. The normality together with the similar slopes appears to confirm the impression from the box plots that the distributions differ by additive shifts.

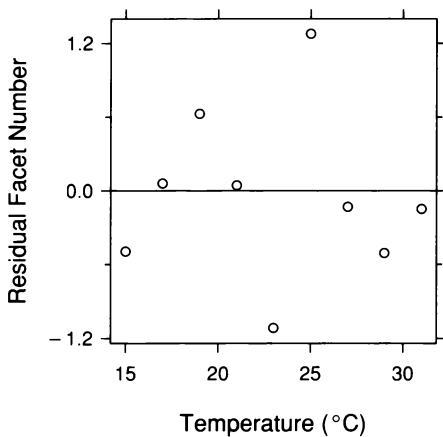


3.48 The conditional distributions of facet number given temperature are compared with the normal distribution by normal q-q plots of the residuals from the means.

Our goal now is to determine how the locations of the distributions of facet number vary with temperature. The box plots in Figure 3.47 suggest that the overall effect is linear. But there is anomalous behavior at 23°C and 25°C; the distribution at 25°C is shifted upward compared to the distribution at 23°C, which runs counter to the trend. The overall trend and the anomaly are probed further by the visualizations in Figures 3.49 and 3.50. The first figure displays both the least-squares fit of a line to the data and the means of the nine distributions. The second figure displays the residuals of the nine means from the least-squares line. Clearly the means at 23°C and 25°C vary by large amounts from the overall pattern.



3.49 The plotting symbols graph the means of the nine conditional distributions of facet number given temperature. The line is the least-squares fit to the 823 measurements of facet number and temperature.



3.50 The deviations of the conditional means from the least-squares line are graphed against temperature.

The scientific question is this. Should we conclude that there is a reproducible anomaly in the dependence of facet number on temperature? Or, is the effect linear, and some aspect of the experiment has caused a result that would not appear in a more carefully controlled experiment? Two pieces of information are important for making the decision. First, a linear decrease of facet number measured on a log scale is expected based on zoological reasoning and other data. The experimenters state:

Seyster ('19) and Krafka ('20) showed that the number of facets decreases with the increase in temperature in bar and ultra-bar races of *Drosophila*. The rate of decrease is proportional to the number of facets within any given stock, so that we are dealing with an instance of 'the compound interest law'.

Second, the values of the factor are nominal values and not necessarily what was achieved. The average temperature of an incubator fluctuates with time, and the temperature at a point in time is not constant throughout the incubator. Here is what the experimenters write about the control of temperature in the experiment:

Errors in the study may arise from fluctuating temperatures. The fluctuations are not the same in all cases; 15°, 17°, 25°, and 27° showed a variation of $\pm 0.5^\circ$. 23° varied as much as 3°. The record of 29° showed a variation from 28.0° to 29.5°. Temperature readings taken of the food by a thermometer graded in tenths of degrees, showed a total variation of 1.4° in the 31° aquarium, and of 1.3° in the 21° aquarium. The 19° aquarium was more unsatisfactory and varied sometimes as much as 5°. In every case, however, an attempt was made to have the proper temperature during the facet reaction period, though this may not have been accomplished because the period may not be the same for the heterozygotes as for the homozygotes for which it has been determined. Nominal temperatures in accompanying tables and figures are to be considered with these remarks in mind.

Note, in particular, the large uncertainty in the temperature at 23°C. The visualization of the data, the zoological reasoning, and the information

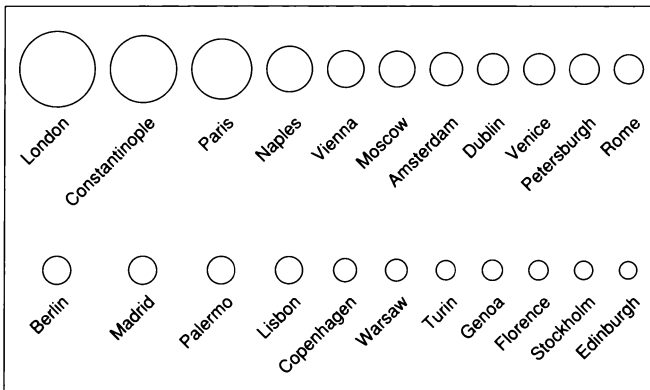
about the experimental protocol suggest the conclusion that there is no meaningful temperature anomaly and that the dependence of facet number on temperature is linear.

Using probabilistic methods to enhance our inductive inference on this matter would require incorporating the uncertainty of the temperature measurements into the analysis, which would be complicated. And attempting to carry out probabilistic methods without accounting for the uncertainty would be uninformative or worse yet, misleading. Surprisingly, R. A. Fisher, the founder of modern statistical theory, made such an attempt. This will be discussed in the final section of the chapter.

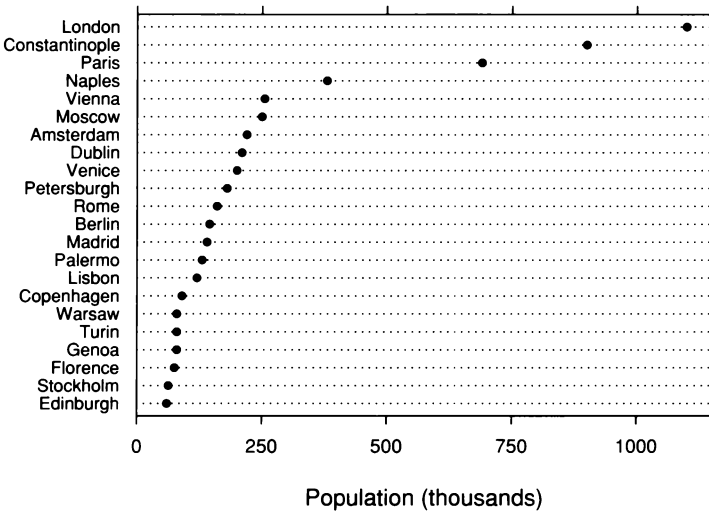
3.9 Transforming Factor and Response

In 1801, William Playfair published his *Statistical Breviary* [67], which contains many displays of economic and demographic data. One display, beautifully reproduced by Tufte [74], graphs the populations of 22 cities by the areas of circles. The graph also contains a table of the populations, so we can compare the data and the areas of the circles.

Figure 3.51 graphs the data by circles whose areas are proportional to those of Playfair. Using area to encode quantitative information is a poor graphical method. Effects that can be readily perceived in other visualizations are often lost in an encoding by area [20]. For example, Figure 3.52 is a dot plot of the populations that conveys the distribution of the data more effectively than the areas of Figure 3.51.



3.51 Circle areas encode the populations of cities at the end of the 1700s.



3.52 A dot plot displays the population data.

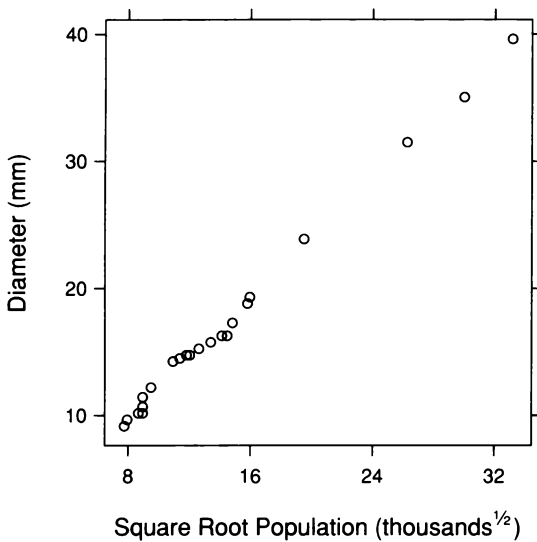
Still, for historical purposes, it is interesting to study Playfair’s areas to see how accurately the circles have been drawn. Such study is inspired by the observation that the circle area for Turin is slightly less than that for Genoa, even though the population values recorded on Playfair’s graph for these cities are equal. Thus the goal is to study how the areas depend on the populations, so area is a response and population is a factor.

Instead of studying population and area directly, we will work on a square root scale. In place of area, we will use diameter. In place of population, we will use square root population. There are two reasons. First, drawn circles are typically rendered by specifying a linear measure such as a radius or a diameter. Thus we might expect that for the diameters, measurement error would be roughly constant. Second, the distributions of the populations and the areas are substantially skewed toward large values, and taking square roots removes much of the asymmetry. Thus we are transforming both the factor and the response to simplify the analysis.

Let d_i be the diameters of the circles in Tufte's reproduction, and let s_i be the square root populations. In Figure 3.53, d_i is graphed against s_i . If the encoding by circle area were exact, the circle areas would be proportional to the populations, and the transformed data would lie along a line of the form

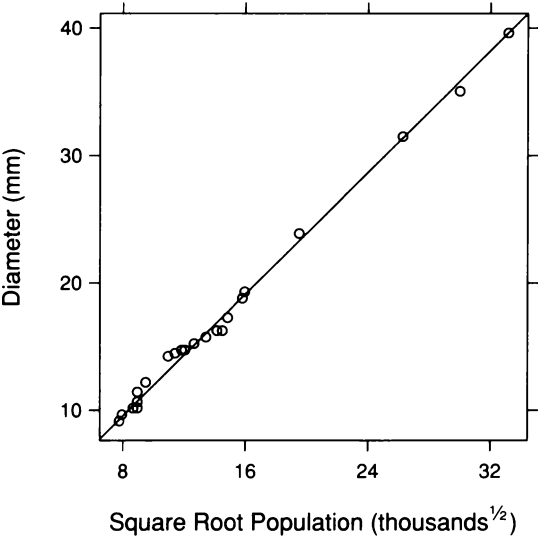
$$d_i = bs_i .$$

This is a linear parametric family with a single parameter b , and an intercept of zero.



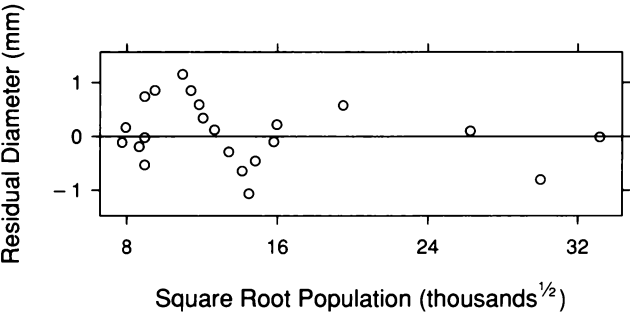
3.53 Circle diameter is graphed against square root population.

Figure 3.54 graphs the data and the least-squares fit of the linear parametric family with zero intercept; the display shows how close the data come to an exact encoding. There is a consistent departure from linearity — an undulation back and forth for those observations with diameters of about 10 mm to 20 mm. However, the magnitude of this undulation is very small compared with the overall variation in the data, so the graph gives us little information about its properties.



3.54 A line has been fitted to the Playfair data by least-squares.

Figure 3.55 is a residual dependence plot. By removing the gross linear effect, the plot magnifies the deviations from a perfect encoding. Now we can perceive the properties of the deviations far more incisively. Their total variation is about ± 1 mm. Furthermore, there is clearly a dependence of the residuals on square root population. This suggests that the deviations are not simply the usual measurement error that would result from laying down a length with a measuring device. Such measurement error should produce a random pattern with no dependence. The true cause of the deviations is likely to remain a mystery without more information about the production of the original graph.



3.55 The residuals from the linear fit to the Playfair data are graphed against square root population.